

Marketing Funnel & Conversion Performance Analysis

1. Project Overview

This project analyzes the marketing campaign performance of a Portuguese banking institution using the Bank Marketing Dataset. The primary objective is to evaluate customer conversion behavior, identify drop-off points within the marketing funnel, and provide actionable recommendations to improve campaign effectiveness.

The analysis focuses on customer demographics, campaign performance, communication channels, and historical campaign outcomes to understand factors influencing term deposit subscriptions.

2. Business Objective

The project aims to answer the following business questions:

- What is the overall conversion rate of the marketing campaign?
 - Where do customers drop off in the conversion funnel?
 - Which communication channels perform best?
 - Which customer segments are more likely to convert?
 - How do previous campaign outcomes influence future conversions?
 - What actions can improve campaign performance?
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3. Dataset Information

Dataset: Bank Marketing Dataset

Source: UCI Machine Learning Repository

Records: 45,211

Features: 17

Target Variable: y (Term Deposit Subscription)

Target Values:

- Yes = Customer subscribed
- No = Customer did not subscribe

The dataset contains customer demographics, financial information, campaign details, and previous marketing outcomes.

4. Methodology

The project was completed using Python, Jupyter Notebook, and Power BI.

Data Preparation

- Loaded and explored the dataset
- Verified data quality
- Checked for missing values
- Reviewed data types and feature distributions
- Created age group bins for customer segmentation

KPI Development

The following KPIs were created:

- Total Leads
- Converted Customers
- Conversion Rate (%)
- Drop-Off Rate (%)

Dashboard Development

A multi-page Power BI dashboard was designed to analyze marketing funnel performance from different perspectives.

5. Dashboard Overview

Page 1: Executive Summary

Provides a high-level overview of campaign performance.

Key Visuals:

- Total Leads
- Converted Customers
- Conversion Rate
- Drop-Off Rate
- Monthly Conversion Performance
- Conversion Rate by Contact Method
- Conversion Rate by Previous Campaign Outcome

Page 2: Campaign Performance Analysis

Evaluates campaign effectiveness.

Key Visuals:

- Conversion Rate by Number of Contacts
- Conversion Rate by Contact Method
- Conversion Rate by Previous Campaign Outcome

Page 3: Customer Insights Analysis

Examines customer characteristics affecting conversion.

Key Visuals:

- Conversion Rate by Job
- Conversion Rate by Education

- Conversion Rate by Marital Status
 - Conversion Rate by Age Group
 - Conversion Rate by Housing Loan Status
 - Conversion Rate by Loan Status
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6. Key Findings

Overall Funnel Performance

- Total Leads: 45,211
- Converted Customers: 5,289
- Conversion Rate: 11.7%
- Drop-Off Rate: 88.3%

Although the campaign generated a large number of leads, a significant proportion did not convert, indicating opportunities for optimization.

Communication Channel Performance

Cellular communication achieved the highest conversion rates compared to telephone and unknown contact methods, making it the most effective communication channel.

Previous Campaign Impact

Customers who responded positively to previous campaigns demonstrated substantially higher conversion rates than other customer groups.

Customer Demographics

- Customers with tertiary education showed stronger conversion performance.
- Conversion behavior varied across occupations.
- Certain age groups exhibited higher conversion rates than others.
- Customers without housing or personal loans generally demonstrated better conversion performance.

Campaign Contact Strategy

Conversion rates declined as the number of customer contacts increased, suggesting diminishing returns from repeated outreach.

7. Business Recommendations

1. Prioritize customers with a history of successful campaign responses.
 2. Increase the use of cellular communication channels.
 3. Focus marketing efforts on high-performing customer segments.
 4. Optimize campaign timing based on historical conversion trends.
 5. Reduce excessive customer contact frequency to avoid diminishing returns.
 6. Develop targeted campaigns for customer groups with higher conversion potential.
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8. Conclusion

This analysis demonstrates how marketing funnel analytics can be used to improve campaign effectiveness and customer targeting strategies. The results highlight the importance of customer segmentation, communication channels, campaign history, and contact frequency in driving conversions.

By leveraging these insights, organizations can improve marketing efficiency, reduce customer drop-off, and achieve higher conversion rates in future campaigns.